CUET_InferX at ImageEval 2026 Shared Tasks: Audio-Visual LoRA Adaptation of Qwen2.5-Omni for Culturally-Grounded Spoken VQA

Md. Ashraful Islam Semon, Jihadul Islam
Department of Computer Science and Engineering
Chittagong University of Engineering and Technology, Bangladesh
{u2104059,u2104080}@student.cuet.ac.bd

Abstract

We present CUET_InferX, our submission to Subtask 1a (Spoken VQA, English-audio) of the ImageEval 2026 shared task on the AynVQA dataset. The task requires predicting the correct answer option solely from an image and a raw audio clip containing both a spoken question and three candidate options. We fine-tune `Qwen2.5-Omni-3B` with LoRA applied exclusively to the language reasoning backbone, keeping both audio and vision encoders frozen, under a single 16 GB Kaggle T4 GPU. Training uses a time-budgeted, class-stratified subsample of 2,055 examples. Our system achieves **91.20%** test accuracy (Balanced Accuracy 87.72%, Macro-F1 80.50%), placing 3rd position, with gains of $+7.20$ accuracy and $+15.95$ Macro-F1 over the zero-shot base. We compare against a cascaded ASR→VLM alternative and zero-shot 3B/7B baselines, and explain with devtest evidence why we submitted the end-to-end fine-tuned model despite lower raw test accuracy than the cascade.

1 Introduction

Spoken VQA over culturally situated imagery requires a system to jointly interpret acoustic content, visual content, and cultural context – a significantly harder joint task than any of these individually. The ImageEval 2026 shared task (Abdaljalil et al., 2026) builds on the AynVQA, a culturally grounded multimodal benchmark covering visual content from 18 Arab countries with spoken questions.

In **Subtask 1a** (English-audio track), a system receives an image I and a single audio waveform A that verbally states a question followed by three candidate options. The output is a single index $y \in \{0, 1, 2\}$; no text transcription is provided . One natural approach is to cascade: transcribe the audio with an ASR system and pass the transcript to a vision-language model. The other is to process audio and image jointly with an omni-modal model.

We initially hypothesized the end-to-end route would be more robust: a transcription error on short, phonetically similar tokens ("*Option 0* . . . *Option 1* . . . *Option 2* . . . ") or on culturally specific terms is unrecoverable downstream. We did not assume this was true – we tested both designs, and our decision to submit the end-to-end system rests on devtest evidence (Section 6.2).

Contributions.

- End-to-end spoken VQA framed as single-token constrained generation over raw audio and pixels, avoiding a cascaded ASR stage.
- LoRA adaptation of only the language reasoning backbone (41.08M parameters, 0.865% of 4.74B total) fitting a 16 GB GPU, with frozen perception towers.
- A class-stratified data-budgeting protocol to avoid option-position bias under a fixed compute budget.
- A four-way ablation with an explicit, evidence-based justification for the submitted configuration.

Detailed implementation information is available in the GitHub repository- `https://github.com/semon87/imageeval-task1a`

2 Related Work

Culturally-Aware Visual Reasoning Standard VQA benchmarks (Antol et al., 2015; Goyal et al., 2018; Hudson and Manning, 2019) predominantly evaluate Western commonsense knowledge. To address geographic and cultural bias (Parcalabescu et al., 2022), the AynVQA and OASIS benchmarks (Abdaljalil et al., 2026; Alam et al., 2025) introduce

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 747–752
October 28-29, 2026 ©2026 Association for Computational Linguistics

visual reasoning grounded in Arab regional heritage and artifacts. Recent evaluations reveal that multimodal architectures frequently suffer from cross-modal instability and severe counterfactual hallucinations when processing culturally situated content (Mousi et al., 2026a,b).

Spoken VQA: Direct vs. Cascaded Systems Spoken VQA requires joint perceptual alignment across acoustic waveforms and visual scenes (Chrupała, 2022). Traditional approaches chain an ASR front-end (Radford et al., 2022) to a vision-language backbone (Bai et al., 2025b), but compounding transcription errors and phonetic information loss degrade reasoning on spoken regional dialects (Mousi et al., 2026a). Native omni-modal transformers (Xu et al., 2025) resolve this by jointly encoding raw audio and visual tokens within a unified transformer architecture.

Parameter-Efficient Adaptation Parameter-Efficient Fine-Tuning (PEFT; Houlsby et al., 2019), notably Low-Rank Adaptation (LoRA; Hu et al., 2022; Dettmers et al., 2023; Zhang et al., 2023), enables resource-constrained task transfer. In multimodal models, selectively adapting the language reasoning core while locking visual and acoustic encoders prevents catastrophic forgetting of foundational representations while optimizing GPU memory utilization (Alayrac et al., 2022; Xu et al., 2025).

3 Task and Dataset

Formulation. Given image I and audio A = [question] $\oplus$ [options 0,1,2], predict $y \in \{0, 1, 2\}$.

Data. Table 1 summarizes the splits. A critical characteristic is the *train/test acoustic domain shift*: all training, dev, and devtest audio is synthetic (voice-cloned), while the blind test set uses human recordings from diverse speakers and conditions.

Metrics. The ranking metric is **Accuracy**; Balanced Accuracy and Macro-F1 are secondary. Since labels denote answer *position* (not semantics), Balanced Accuracy specifically surfaces positional bias that raw accuracy can mask under class imbalance.

4 System Architecture

Figure 1 gives an overview. We describe each component in turn.

4.1 Base Model: Qwen2.5-Omni-3B

`Qwen2.5-Omni-3B` (Xu et al., 2025) consists of three components: (i) a 32-layer convolutional/SDPA **audio encoder**; (ii) a 32-layer **vision encoder** with 3D patch embedding and patch merger; and (iii) a 36-layer decoder-only **Thinker** language core for cross-modal reasoning. Since the task requires text output rather than synthesized speech, we disable the Talker head via `disable_talker()`.

Split	N	Audio	Labels
Train	3,000	Synthetic	Yes
Dev	500	Synthetic	Yes
Devtest	500	Synthetic	Yes
Test	1,000	Human	Post-competition

Table 1: Dataset splits and acoustic domain.

4.2 Parameter-Efficient Fine-Tuning

We freeze both perception towers entirely:

$$\nabla_{\theta_{\text{audio}}} \mathcal{L} = \mathbf{0}, \qquad \nabla_{\phi_{\text{vis}}} \mathcal{L} = \mathbf{0}$$

and inject LoRA (Hu et al., 2022) into all attention projections (W_q, W_k, W_v, W_o) and MLP gates ($W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}$) of the Thinker, with rank r=16, α=32, dropout p=0.05. This gives 41.08M trainable parameters out of 4.74B total (0.865%).

The rationale for freezing the encoders is twofold: it limits catastrophic forgetting of pretrained perceptual representations, and it substantially reduces GPU memory pressure. The perception towers were already trained on large-scale audio/visual corpora; the adaptation gap for this task lies in how the language backbone reasons over their outputs under a constrained output format, not in the encoders' feature extraction.

4.3 Memory Optimizations

To stay within 16 GB: (i) SDPA attention avoids the $\mathcal{O}(L^2)$ memory cost of eager attention on long multimodal sequences; (ii) vision input is capped at 512×512 pixels, limiting vision token count; (iii) non-reentrant gradient checkpointing is enabled via `enable_input_require_grads()`.

4.4 Prompting and Loss Masking

The input is formatted as a three-turn conversation: a system turn instructing the model to output only a single digit; a user turn containing the audio and

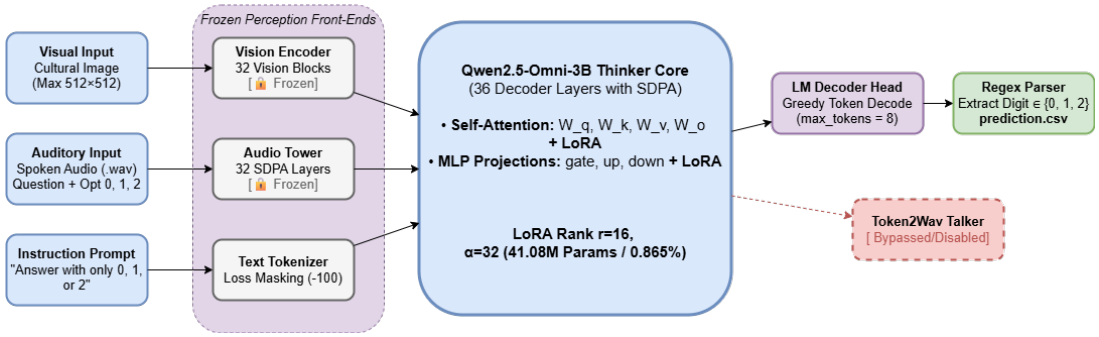


Figure 1: CUET_InferX pipeline: frozen vision and audio encoders feed the LoRA-adapted Thinker core. The speech-synthesis Talker head is disabled since only a text digit is required at output. Loss is computed on the assistant-turn digit token only.

image with the instruction "Answer with only 0, 1, or 2"; and an assistant turn targeting the gold digit. Cross-entropy loss is restricted to the assistant token only (all other tokens masked with -100). The exact system prompt is in Appendix A.

5 Experimental Setup

Data budgeting. Due to GPU and time constraints, training was conducted on a single Tesla T4 with a maximum training budget of approximately 7 hours. With an average processing time of $t_{\text{step}} \approx 12.26$s per sample, this budget corresponds to:

$$N_{\text{budget}} = \left\lfloor \frac{7 \times 3600}{12.26} \right\rfloor \approx 2{,}055$$

Accordingly, we used 2,055 training samples. To maintain a balanced class distribution and avoid inheriting any option-position imbalance from the original training set, we selected exactly 685 samples from each class and globally shuffled the resulting subset before training.

Hyperparameters. All runs use bfloat16, seed 42, AdamW ($\eta = 10^{-4}$), 1 epoch (256 optimizer steps), effective batch size 8 (per-device 1, gradient accumulation 8), gradient clipping 1.0, and 512×512 image resolution. Training completed in 482.8 minutes. Full details are in Appendix A.

Inference. We merge the LoRA adapter (`merge_and_unload()`), decode greedily (`do_sample=False`, `max_new_tokens=8`), and parse the first valid digit via regex. A uniform random fallback exists for the case where no valid digit appears (which never occurred in practice).

Ablation configurations. To test our end-to-end design choice with evidence rather than assumption, we evaluate three additional configurations on both the 500-item devtest split (where we have labels) and the blind test set: (i) **Zero-shot Qwen2.5-Omni-3B** ; (ii) **Zero-shot Qwen2.5-Omni-7B**; (iii) **Cascaded ASR→VLM**: faster-whisper large-v3 (Radford et al., 2022) for transcription, feeding the transcript and image to Qwen3-VL-8B-Instruct (Bai et al., 2025a) with an equivalent single-digit prompt. We could not LoRA fine-tune the 7B backbone due to persistent CUDA OOM errors on 16 GB hardware even at minimal batch/resolution settings.

6 Results and Analysis

6.1 Official Leaderboard

#	Team	Acc. ↑	Bal. Acc.	Macro-F1
1	am2401295	0.9620	0.9332	0.9081
2	NYUAD	0.9610	0.9193	0.8718
3	**CUET_InferX**	**0.9120**	**0.8772**	**0.8050**

Table 2: Official Subtask 1a leaderboard (blind human-recorded test set).

6.2 Ablation: Why the Fine-Tuned End-to-End 3B Model?

Table 3 compares all four configurations on devtest and the blind test set.

Fine-tuning vs. zero-shot (same backbone). LoRA adaptation over the 3B base yields +7.20 accuracy, +15.95 Macro-F1, and +15.10 Balanced Accuracy points on test – the only variable between these two rows is the adapter, making this the cleanest evidence in the paper.

Fine-tuned 3B vs. zero-shot 7B (post-competition). The zero-shot 7B model

System	DevT	Acc.	BAcc.	F1
Omni-3B (ZS)	–	0.8400	0.7262	0.6455
Omni-7B (ZS)	–	0.9340	0.8618	0.8212
ASR→VLM	0.9160	**0.9370**	0.8781	**0.8131**
Omni-3B+LoRA	**0.9360**	0.9120	**0.8772**	0.8050

Table 3: Devtest accuracy (DevT) and blind-test performance. Acc., BAcc., and F1 denote accuracy, balanced accuracy, and macro-F1. ZS denotes zero-shot.

outperforms our system by 2.2 accuracy points using ∼58% more parameters and no task-specific data. We read this as a compute-accessibility result: 41M trainable parameters on a 16 GB GPU recover most of what 2.6B additional parameters provide, with LoRA fine-tuning of the 7B backbone being the most promising direction we could not pursue.

Fine-tuned 3B vs. cascaded ASR→VLM. This comparison directly determined our submission choice. On devtest – the only labeled split available before submission – our model scored 93.60% vs. the cascade's 91.60%, so we submitted the end-to-end system. On the blind test set the relationship inverted (91.20% vs. 93.70%), which we attribute to the synthetic-to-human acoustic domain shift: faster-whisper large-v3, trained on diverse human speech, transfers more gracefully than our frozen audio encoder, which saw only synthetic audio during fine-tuning. The devtest gap is consistent with cascaded fragility on culturally specific vocabulary, where ASR mis-transcriptions propagate uncorrected to the downstream VLM.

Takeaway. Based on devtest evidence, the end-to-end fine-tuned 3B model was the right call at submission time. The cascade's test-set advantage emerged only from an acoustic domain shift that was unobservable before the blind set was released.

6.3 Class-Level Analysis

Label	True	Pred	P	R	F1
0	539	479	0.9812	0.8720	0.9234
1	437	478	0.8849	0.9680	0.9246
2	24	43	0.4419	0.7917	0.5672
Macro	1000	1000	0.7693	0.8772	0.8050

Table 4: Per-class test report. Micro-F1: 0.9120; Weighted-F1: 0.9154; κ: 0.8346; MCC: 0.8392.

The test distribution is heavily skewed – Option 0 (53.9%), Option 1 (43.7%), Option 2 (2.4%) – which is the primary driver of the Macro-F1/Balanced Accuracy gap. Option 2's low F1 (0.5672) drags down the macro average despite reasonable recall (0.7917); Balanced Accuracy is less affected since it only weights per-class recall. The model over-predicts Option 2 (43 vs. 24 gold) and under-predicts Option 0 (479 vs. 539), a residual bias from stratified training that upsampled Option 2 relative to its natural test frequency. Figure 2 shows the full confusion matrix.

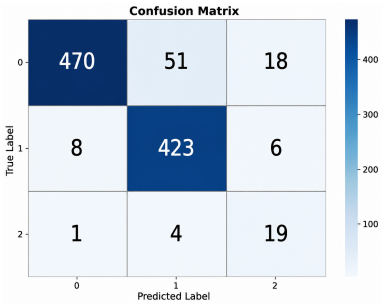


Figure 2: Confusion matrix of test predictions.

Notably, greedy decoding produced a valid digit on all 1,000 test items with zero fallback invocations, confirming the adapter reliably enforces the constrained output format even under human-audio conditions it was not trained on.

7 Conclusion and Limitations

We fine-tuned `Qwen2.5-Omni-3B` with LoRA on its language backbone – 41.08M trainable parameters, single 16 GB GPU, class-stratified subsample of 2,055 examples – for end-to-end spoken VQA, reaching 91.20% test accuracy and achieved 3rd position. Our ablation study shows the submitted design was the right call given devtest evidence, even though a cascade and a larger zero-shot model both outscored it on the blind test set. Future directions include LoRA fine-tuning of the 7B backbone, audio-encoder adaptation for human speech, and country/category-stratified comparison of end-to-end and cascaded pipelines.

Limitations. We subsampled training from 3,000 to 2,055 items under session time constraints. The 7B backbone comparison is zero-shot only due to memory limits. The cascade's test-set advantage stems from the acoustic domain shift, which the frozen audio encoder could not compensate for. We evaluated only the English-audio track.

References

Samir Abdaljalil, Hunzalah Hassan Bhatti, Ahlam Bashiti, Farina Amir, Md Arid Hasan, Basel Mousi, Nadir Durrani, Fahim Dalvi, Zien Sheikh Ali, Erchin Serpedin, Hasan Kurban, Mustafa Jarrar, Shammur Absar Chowdhury, and Firoj Alam. 2026. ImageEval 2026: Culturally grounded Arabic multimodal evaluation. In *Proceedings of the Fourth Arabic Natural Language Processing Conference: Shared Tasks*, Budapest, Hungary. Association for Computational Linguistics.

Firoj Alam, Ali Ezzat Shahroor, Md. Arid Hasan, Zien Sheikh Ali, Hunzalah Hassan Bhatti, Mohamed Bayan Kmainasi, Shammur Absar Chowdhury, Basel Mousi, Fahim Dalvi, Nadir Durrani, and Natasa Milic-Frayling. 2025. OASIS: A multilingual and multimodal dataset for culturally grounded spoken visual QA. *arXiv preprint arXiv:2510.06371*.

Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, et al. 2022. Flamingo: A visual language model for few-shot learning. *Proceedings of NeurIPS*, 35:23716–23736.

Stanislaw Antol, Aishwarya Agrawal, Jiasen Lu, Margaret Mitchell, Dhruv Batra, C. Lawrence Zitnick, and Devi Parikh. 2015. Vqa: Visual question answering. In *2015 IEEE International Conference on Computer Vision (ICCV)*, page 2425–2433. IEEE.

Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, Wenbin Ge, Zhifang Guo, Qidong Huang, Jie Huang, Fei Huang, Binyuan Hui, Shutong Jiang, Zhaohai Li, Mingsheng Li, and 45 others. 2025a. Qwen3-vl technical report. *arXiv preprint arXiv:2511.21631*.

Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, and 8 others. 2025b. Qwen2.5-vl technical report. *arXiv preprint arXiv:2502.13923*.

Grzegorz Chrupała. 2022. Visually grounded models of spoken language: A survey of datasets, architectures and evaluation techniques. *Journal of Artificial Intelligence Research*, 73:673–707.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient finetuning of quantized LLMs. In *Thirty-seventh Conference on Neural Information Processing Systems*.

Yash Goyal, Tejas Khot, Aishwarya Agrawal, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. 2018. Making the v in vqa matter: Elevating the role of image understanding in visual question answering. *International Journal of Computer Vision*, 127(4):398–414.

Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin De Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. 2019. Parameter-efficient transfer learning for NLP. In *Proceedings of ICML*, pages 2790–2799.

Edward J Hu, yelong shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

Drew A. Hudson and Christopher D. Manning. 2019. Gqa: A new dataset for real-world visual reasoning and compositional question answering. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, page 6693–6702. IEEE.

Basel Mousi, Fahim Dalvi, Shammur Chowdhury, Firoj Alam, and Nadir Durrani. 2026a. Said aloud, read different: Cross-modal instability in multimodal models. In *Proceedings of Interspeech 2026*, Sydney, Australia. Accepted.

Basel Mousi, Fahim Dalvi, Shammur Absar Chowdhury, Firoj Alam, and Nadir Durrani. 2026b. Once correct, still wrong: Counterfactual hallucination in multilingual vision-language models. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 4763–4788, San Diego, California, United States. Association for Computational Linguistics.

Letitia Parcalabescu, Michele Cafagna, Lilitta Muradjan, Anette Frank, Iacer Calixto, and Albert Gatt. 2022. VALSE: A task-independent benchmark for vision and language models centered on linguistic phenomena. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Dublin, Ireland. Association for Computational Linguistics.

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2022. Robust speech recognition via large-scale weak supervision. *arXiv preprint*.

Jin Xu, Zhifang Guo, Jinzheng He, Hangrui Hu, Ting He, Shuai Bai, Keqin Chen, Jialin Wang, Yang Fan, Kai Dang, Bin Zhang, Xiong Wang, Yunfei Chu, and Junyang Lin. 2025. Qwen2.5-omni technical report. *arXiv preprint arXiv:2503.20215*.

Renrui Zhang, Jiaming Han, Aojun Zhou, Xiangfei Hu, Shilin Yan, Pan Lu, Hongsheng Li, Peng Gao, and Yu Qiao. 2023. LLaMA-Adapter: Efficient fine-tuning of language models with prefix-tuning and dynamic adaptation. *arXiv preprint arXiv:2303.16199*.

A Reproducibility Details

A.1 Software

`transformers==4.52.3`, `peft==0.14.0`, `accelerate==1.4.0`,

`torch==2.6.0+cu124`, `qwen-omni-utils`, `faster-whisper`. `PYTORCH_CUDA_ALLOC_CONF=''expandable_segments:True''` was set prior to CUDA init to reduce fragmentation; `torchao` was uninstalled to prevent quantization conflicts.

A.2 System Prompt

```
"You are given an image and an
audio clip.  The audio asks a
question about the image and
then reads three answer options:
Option 0, Option 1, Option 2.
Listen carefully and look at the
image, then reply with ONLY a
single digit:  0, 1, or 2 ---
the index of the correct option.
No other words."
```

A.3 Hyperparameters

Hyperparameter	**Value**
Base model	Qwen2.5-Omni-3B
LoRA rank r	16
LoRA α	32
LoRA dropout	0.05
Target modules	q,k,v,o,gate,up,down_proj
Trainable params	41.08M (0.865% of 4.74B)
Training samples	2,055 (685/class, stratified)
Epochs	1 (256 optimizer steps)
Effective batch size	8 (per-device 1, accum. 8)
Learning rate	1×10^{-4}
Optimizer	AdamW
Max image pixels	512×512
Precision	bfloat16
Attention	SDPA
GPU	1$\times$ Tesla T4 (16 GB)
Training time	482.8 min
Seed	42

Table 5: Training configuration.